8-MONTH COMPLETE MASTERY PROGRAM

DATA SCIENCE & MACHINE LEARNING

Python · Data Analytics · Machine Learning · Deep Learning · Generative AI
— a complete, structured journey from zero to job-ready Data Scientist.

8 Months	4 Courses	10+ Projects	Live Sessions	100% Support
Full Duration	Complete Path	Hands-On	Expert-Led	Placement

YOUR 8-MONTH LEARNING ROADMAP

Each course builds on the last — a complete, structured progression from zero to job-ready.

01	→	02	→	03	→	04	→	■
Python Fundamentals		Data Analytics		Machine Learning		Deep Learning		Generative AI
6 Weeks		2 Months		3 Months		2 Months		Bonus

DETAILED COURSE BREAKDOWN

Every topic covered — structured, industry-aligned, and project-driven.

COURSE
01
6 WEEKS

Python Programming Fundamentals

Prerequisite: **None — absolute beginners welcome**

Master the complete Python language from scratch. Every session combines theory with live coding demos and hands-on lab exercises. You will write and debug real programs from Week 1.

- Python setup, IDEs & Jupyter Notebooks
- Data Types: int, float, string, boolean
- Variables, Operators & Type Casting
- Decision making: if / elif / else
- Loops: for, while, break, continue
- Functions, Arguments & Return Values
- Lambda Expressions & Higher-order Functions
- Data Structures: Lists, Tuples, Sets, Dicts
- Exception Handling & Custom Exceptions
- File I/O: read, write, CSV & JSON
- Modules, Packages & pip ecosystem
- OOP: Classes, Objects & Methods
- Inheritance, Polymorphism & Encapsulation
- Iterators, Generators & Scopes · 2 Projects

■ OUTCOME ***Write clean, efficient Python programs — tackle real-world problems with professional-grade code***

COURSE

02

2 MONTHS

Data Analytics with Python

Prerequisite: **Python Fundamentals (Course 01)**

Become a confident data analyst using NumPy, Pandas, Matplotlib, and Seaborn. Work with real-world datasets across numerical, categorical, text, and image formats to extract meaningful business insights.

- NumPy: N-d arrays, broadcasting & math ops
- Pandas: DataFrames, Series & multi-indexing
- Data loading: CSV, Excel, JSON, SQL
- Data Assessment & Profiling
- Data Cleaning: nulls, duplicates, outliers
- Data Transformation & Feature Engineering
- Statistical Analysis & Descriptive Stats
- Matplotlib & Seaborn: charts & heatmaps
- Distribution, Correlation & Pair plots
- SciPy: scientific computing & distances
- Working on Categorical & Text data
- RegEx & Text Pre-processing
- Image Data basics with OpenCV
- Dimensionality Reduction · 2 Projects

OUTCOME ***Load, clean, analyse, and visualise any dataset — deliver data-driven insights like a senior analyst***

COURSE

03

3 MONTHS

Advanced Machine Learning

Prerequisite: **Data Analytics (Course 02)**

Master the mathematics and implementation of Machine Learning. From linear models to ensemble methods — build, evaluate, and tune models that solve real business problems using supervised, unsupervised, and reinforcement learning.

- Intro to ML pipeline & workflow
- Linear Regression & Gradient Descent
- Logistic Regression & Bayesian Stats
- Model Evaluation: Accuracy, F1, ROC-AUC
- K-Nearest Neighbors & Decision Trees
- Random Forests & Boosting (XGBoost)
- Support Vector Machines (SVM)
- Ensemble Methods & Stacking
- K-Means, DBSCAN & Agglomerative Clustering
- Dimensionality Reduction: PCA, t-SNE
- Reinforcement Learning fundamentals
- Neural Networks: architecture & backprop
- Feature Selection & Hyperparameter Tuning
- Text & Image ML examples · 3 Projects

OUTCOME ***Build, tune, and deploy ML models that solve classification, regression, and clustering problems at scale***

COURSE

04

2 MONTHS

Deep Learning & Neural Networks

Prerequisite: **Machine Learning (Course 03)**

Go beyond classical ML — train powerful deep neural networks used by Google, Meta, and OpenAI. Implement state-of-the-art architectures for computer vision, sequential data, and generative tasks.

- Deep Neural Networks: layers & activations
- Backpropagation & Optimisers (Adam, SGD)
- TensorFlow & Keras — hands-on training
- CNN: LeNet, AlexNet, VGG architectures
- ResNet & Skip Connections
- Image Classification & Object Detection
- Data Augmentation & Regularisation
- Sequence Modelling: RNN & Vanishing Gradient
- LSTM & Gated Recurrent Units (GRU)
- Transfer Learning & Fine-tuning pretrained models
- Autoencoders & Representation Learning
- Generative Adversarial Networks (GANs)
- Model Deployment with Flask / FastAPI
- GPU batch training · 2 Projects

OUTCOME ***Design and train deep neural networks for vision, NLP, and generative tasks — deploy production AI models***

COURSE

BONUS

Generative AI — The Future of Data Science

Prerequisite: **Any course above — recommended after Course 03 or 04**

The data science landscape has changed forever. Master LLMs, build intelligent pipelines with LangChain, and use AI to supercharge your workflows. Graduate as a modern, AI-fluent data scientist.

- How LLMs work: GPT-4, Gemini, Claude
- OpenAI API & ChatGPT in Python
- Prompt Engineering for data scientists
- LangChain: building intelligent AI pipelines
- RAG (Retrieval-Augmented Generation)
- Vector databases: Pinecone, ChromaDB
- Fine-tuning LLMs on custom datasets
- AI-powered EDA & automated reporting
- GitHub Copilot for DS productivity
- Building AI chatbots for data querying
- Hugging Face models & inference APIs
- Responsible AI, bias detection & ethics

OUTCOME ***Combine ML/DL with Generative AI — build next-gen intelligent data products that command premium salaries***

LIBRARIES & TOOLS YOU WILL MASTER

15 industry-standard tools — used daily at the world's top data science & AI companies.

Python 3.x Core Language	**NumPy** Numerical Computing	**Pandas** DataFrames & EDA	**Matplotlib** Charting	**Seaborn** Stat Plots
Scikit-Learn ML Models	**TensorFlow** Deep Learning	**Keras** Neural Nets	**OpenCV** Image Processing	**NLTK / spaCy** NLP
Jupyter Notebooks	**GitHub** Version Control	**Flask / FastAPI** Deployment	**ChatGPT API** Generative AI	**LangChain** AI Pipelines

CAREER ROLES YOU CAN TARGET

- **Data Analyst** — After Course 02
- **ML Engineer** — After Course 03
- **Deep Learning Engineer** — After Course 04
- **AI / GenAI Developer** — After Bonus
- **Data Scientist** — Full 8-Month Path
- **Research Analyst** — After Course 03

WHO IS THIS FOR?

- **Fresh Graduates**
  BCA, B.Tech, BSc, MBA, BBA — any stream
- **Career Switchers**
  Zero coding background? Start from Course 01
- **Working Professionals**
  Add data skills without quitting your job
- **Analysts & Executives**
  Level up from Excel to Python & ML
- **Entrepreneurs**
  Build intelligent, data-driven products

WHY CODING SHARKS?

- **Industry Expert Mentors**
  Sessions led by working DS & AI engineers
- **Project-Based Learning**
  10+ real-world projects across all courses
- **Full Placement Support**
  Resume, LinkedIn, mock interviews, campus drives
- **Industry Certificate**
  Recognised completion certificate on graduation

Ready to Become a Data Scientist?

Call / WhatsApp
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